Trevecca Event Attendance Analysis

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Business Question

What factors most influence Trevecca students' attendance to on-campus events and how can we use that to increase participation?

- Analyze attendance patterns across recent Trevecca campus events
- Identify key factors influencing attendance using regression modeling
- Survey and evaluate student communication channels (email, social media, word-of-mouth, etc.)
- Determine which event features matter most (food, merch, friends going, interest match)
- Recommend 2–3 actionable strategies to increase participation and improve event communication

Agenda

01 → **Data Preparation**

02 → **Modeling Process**

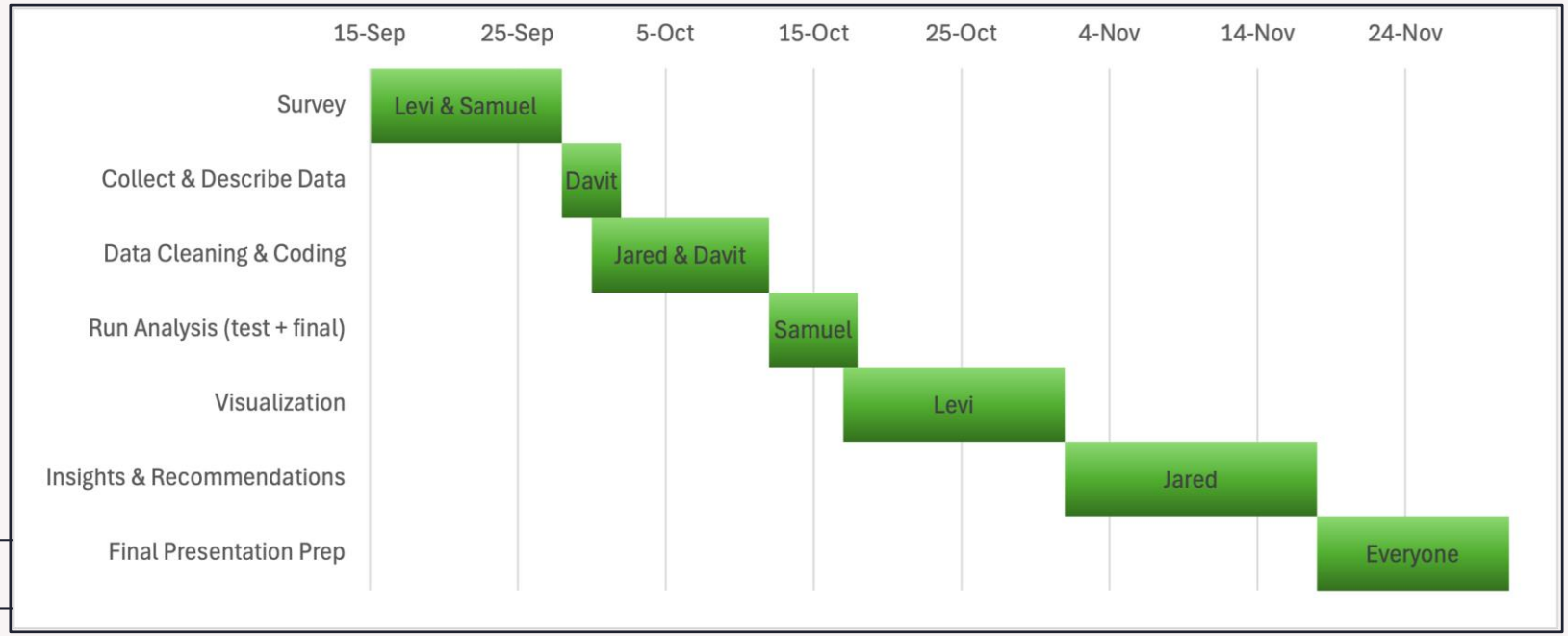
03 → **Evaluation of Results**

04 → **Insights**

05 → **Recommendations**

06 → **Conclusion**

Gantt Chart


15-Sep
25-Sep
5-Oct
15-Oct
25-Oct
4-Nov
14-Nov
24-Nov
Survey
Levi & Samuel
Collect & Describe Data
Davit
Data Cleaning & Coding
Jared & Davit
Run Analysis (test + final)
Samuel
Visualization
Levi
Insights & Recommendations
Jared
Final Presentation Prep
Everyone

Data Sources

Involve Attendance Data (event check-ins & participation trends)

Demographics Data (class year, major, and residential status)

Student Survey (65 responses to 10 categorical questions)

Data Preparation

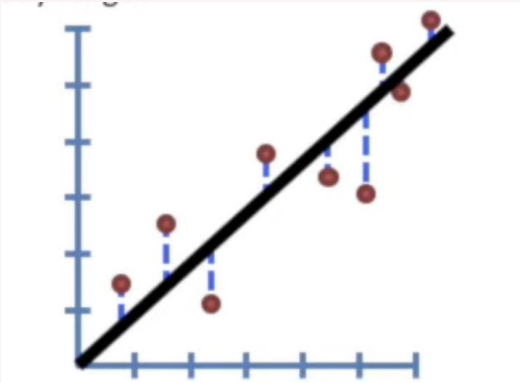
Survey Questions

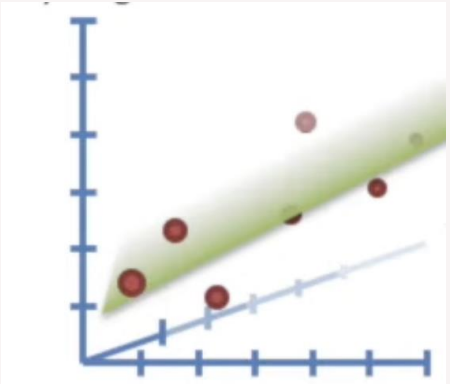
1. What year are you at Trevecca?
2. Do you live on campus?
3. How often do you attend campus events?
4. What types of events do you enjoy the most?
5. What types events do you go to most often?
6. How do you hear about campus events?
7. What gets you excited for the events?
8. What motivates you to attend events?
9. How well do event themes match your interests?
10. How often do you attend events during the semester?

Regression Setup

Why Regression?

- Predicts a continuous attendance score (0–3)
- Handles multiple predictors at the same time
- Quantifies how each factor increases or decreases attendance
- Produces interpretable coefficients for decision-making
- Helps identify which survey factors influence attendance





Excel Preparation

1. **Converted Categorical Answers to Numeric Values.**
   - Many survey questions were text, excel can't run regression text, so all categories were converted into numbers.
2. **Wrote Formulas to Automate the Encoding**
   - Used formulas like Switch(), IFS() or IF() statements to convert each response automatically.
3. **Prepared Clean Columns for Regression**
   - Placed all independent and dependent variables in a right column.

event_type_num	interest_match_num	attendance_score
1	4	1
1	3	1
4	2	1
1	3	0
2	2	2
1	2	1
1	4	0
3	3	0
2	4	1
2	3	1
1	3	1
1	4	3
1	3	1
3	3	0

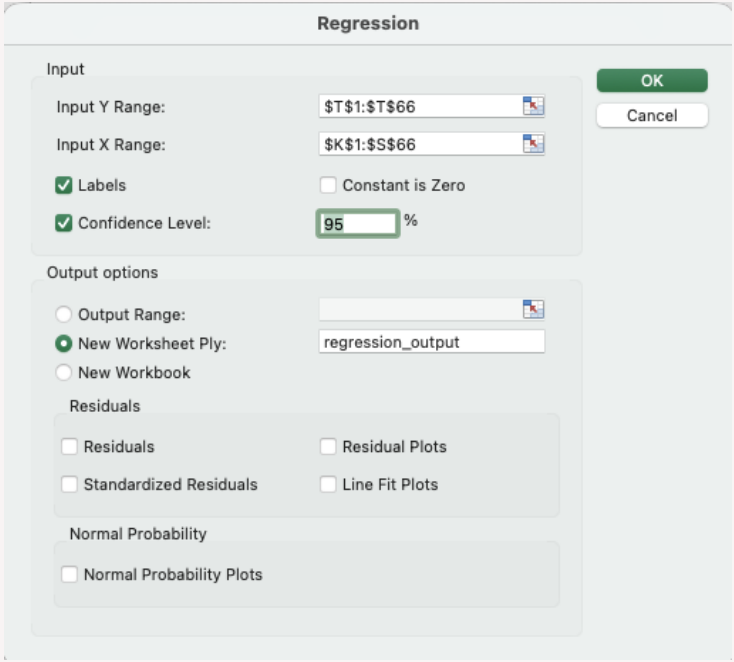
Excel Modeling Overview

- Excel uses Ordinary Least Squares (OLS) regression
- Provides key metrics: R^2, Adjusted R^2, p-values, F-test, Standard Error
- Used to evaluate overall model fit and statistical significance
- Forms the official statistical foundation of our analysis

Regression Setup in Excel

Dependent Variable (Y):
- Attendance Score

Independent Variables (X):
- Year level
- On-campus status
- Awareness count
- Event type preference
- Free food importance
- Merch importance
- Friends influence
- Schedule conflicts
- Interest match



Preparation Steps:
- Converted all categorical answers to numeric values
- Cleaned data for duplicates and missing values
- Ensured columns were contiguous for Excel regression

Excel Model Output

Model shows a moderate fit**,** Some predictors contribute, but attendance is influenced by many factors. This confirms why we used Python to explore predictor importance

Excel Model Output (OLS Results)

- **R^2** = 0.2269 → model explains ~23% of attendance behavior
- **Adjusted R^2** = 0.1004 → model improves after accounting for predictors
- **Significance F** ≈ 0.090 → borderline significance
- **Standard Error** ≈ 0.99

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0,476384957
R Square	0,226942627
Adjusted R Square	0,100442329
Standard Error	0,993437748
Observations	65

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	9	15,93486383	1,770540425	1,794008642	0,090263047
Residual	55	54,28052079	0,98691856		
Total	64	70,21538462			

ANOVA Results

ANOVA tests whether the regression model is statistically meaningful
F = 1.79
Significance F ≈ 0.090

- Model is borderline significant
- Indicates some predictive power, but not strong enough to meet the strict 0.05 threshold
- Expected for behavioral survey data with small sample size (n = 64)

ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regression	9	15,9349	1,77054	1,794	0,090263047
Residual	55	54,2805	0,98692		
Total	64	70,2154			

Why Excel Alone Isn’t Enough

- Excel cannot properly encode categorical variables
- OneHotEncoding is required for:
  - Event type
  - Friend influence
  - Schedule conflicts
  - Interest match

- Excel coefficients become misleading when categories stay as numbers

- Excel is excellent for model fit, not predictor importance

- Python provides the true feature importance using modern preprocessing

Python Regression Setup

Python Libraries

pandas

NumPy

scikit learn

Python Preparation

- Convert to Numeric Scale

“How Often do you attend events?”
- Never = 0

- OneHotEncoding for All Categorical Survey Questions

Applied OneHotEncoder() to every categorical column.
This created binary variables like:
Q1_Freshman = 1 if the student is a Freshman, else 0

WordOfMouth	Posters
0	1
1	0
1	1

- Split into X (Predictors) and Y (Target)

```
X = df.drop(columns=["Q10. How often do you attend events during the semester?", "attendance_score"])
Y = df["attendance_score"]
```

TRAINING, TESTING & RESULTS

Pipeline

1. PreProcessor (ColumnTransformer)

```
preprocessor = ColumnTransformer(
    transformers=[
        ("cat", OneHotEncoder(handle_unknown="ignore"),
categorical_cols)
    ],
    remainder="passthrough"
)
```

2. Linear Regression Model

```
model = Pipeline(steps=[
    ("preprocess", preprocessor),
    ("regressor", LinearRegression())
])
```

Train / Test & Fit Model

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

model.fit(X_train, y_train)
```

- We split our data into Training Data (75%) and into Testing Data (25%)
- We fit the data to actually run the regression after the preprocessing.

Top Predictors

Top Predictors of Student Event Attendance

-Event Type Preference (Q8)

-Friends Going (Q5)

-Importance of Free Food (Q6)

- **Schedule**
- **Event Themes**
- **Year at Trevecca**

Excel (OLS Regression) — Best for Model Fit

Good for:

- R^2, p-values, Significance F
- Stable results with small datasets
- Uses all data at once

Not good for:

- Categorical variables (treats categories as numbers → misleading)

Python Regression — Best for Predictor Importance

Good for:

- Handling categories correctly (OneHotEncoding)
- Finding which factors actually matter
- Comparing many predictors at once

Not good for:

- R^2 with small datasets (because of train/test split)

Final Rule

- **Use Excel** to judge model quality (R^2, p-values).
- **Use Python** to understand which predictors matter most.

Interpretation

Top Predictors of Student Event Attendance

Event Type Preference

→ Students who prefer certain types of events (especially worship, social, sports) attend more.

Friends Going

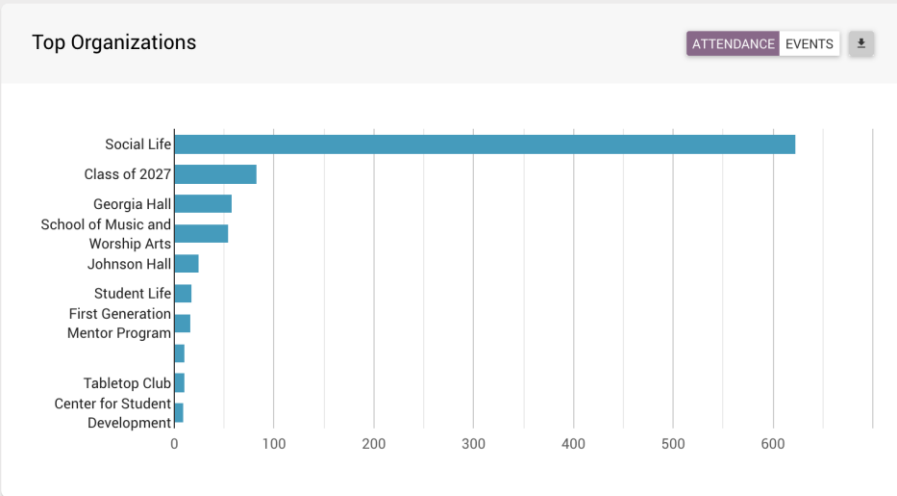
→ Social influence is extremely strong; students attend if their friend groups attend.

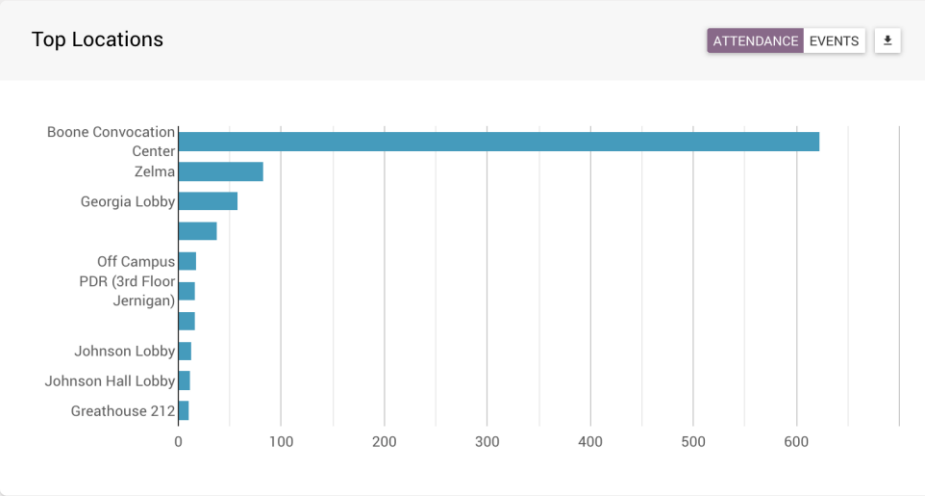
Importance of Free Food

→ Food is a major draw and improves attendance.

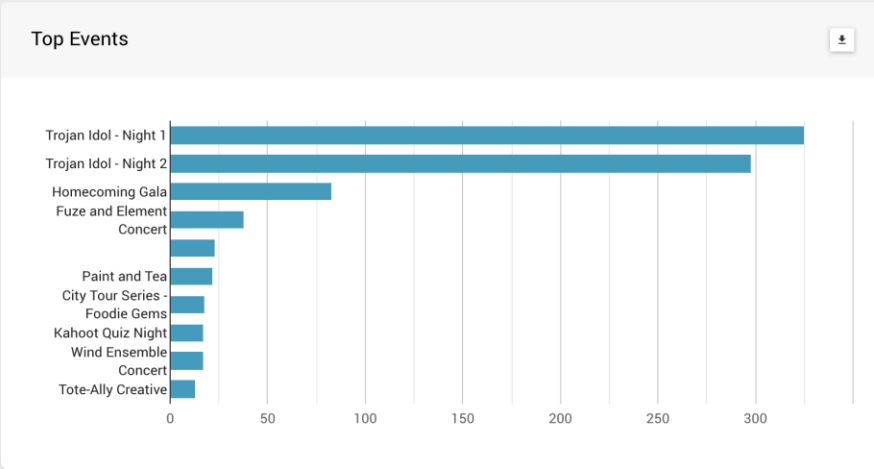
- $\mathbf{R^2}$ = 0.2269 → model explains ~23% of attendance behavior
- 23% of the attendance is affected by the predictors that we have found.

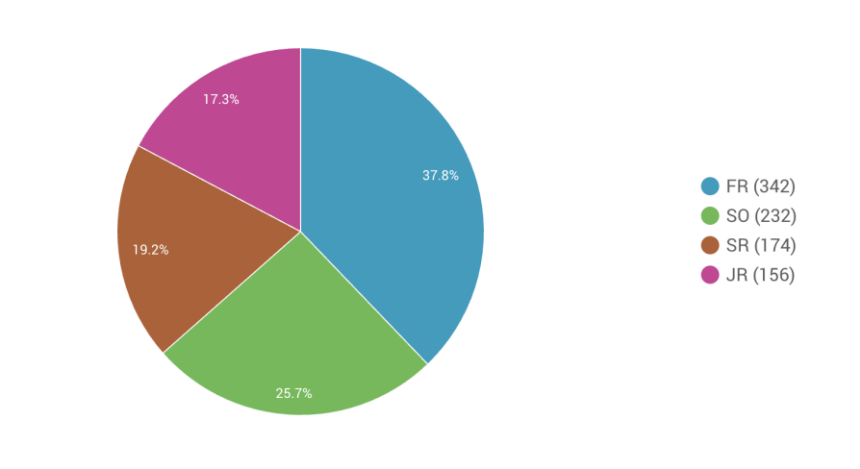
Alignment with CSVs





Alignment with CSVs





Recommendations

- Hold events that align with **successful events** from the past (Trojan Idol, Homecoming Gala, Boonearoo). Marketing through groups, mentors, classes, etc, that align with the themes of events.
- Prioritize **free food** and giveaways at events. Ensuring that this is also part of marketing scheme (Putting that first in a poster).
- Promote events through **group invitations** which share interests. RA group chats, clubs, sports teams. When student ambassadors share, their group will all follow.

Conclusion

- These predictors highlight where Trevecca can make the biggest impact with minimal changes.
- Prioritizing student-preferred event types will naturally increase attendance.
- Encouraging group attendance taps into strong social influence patterns on campus.
- Consistently offering food or small giveaways creates reliable incentives for students to show up.

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THANK YOU